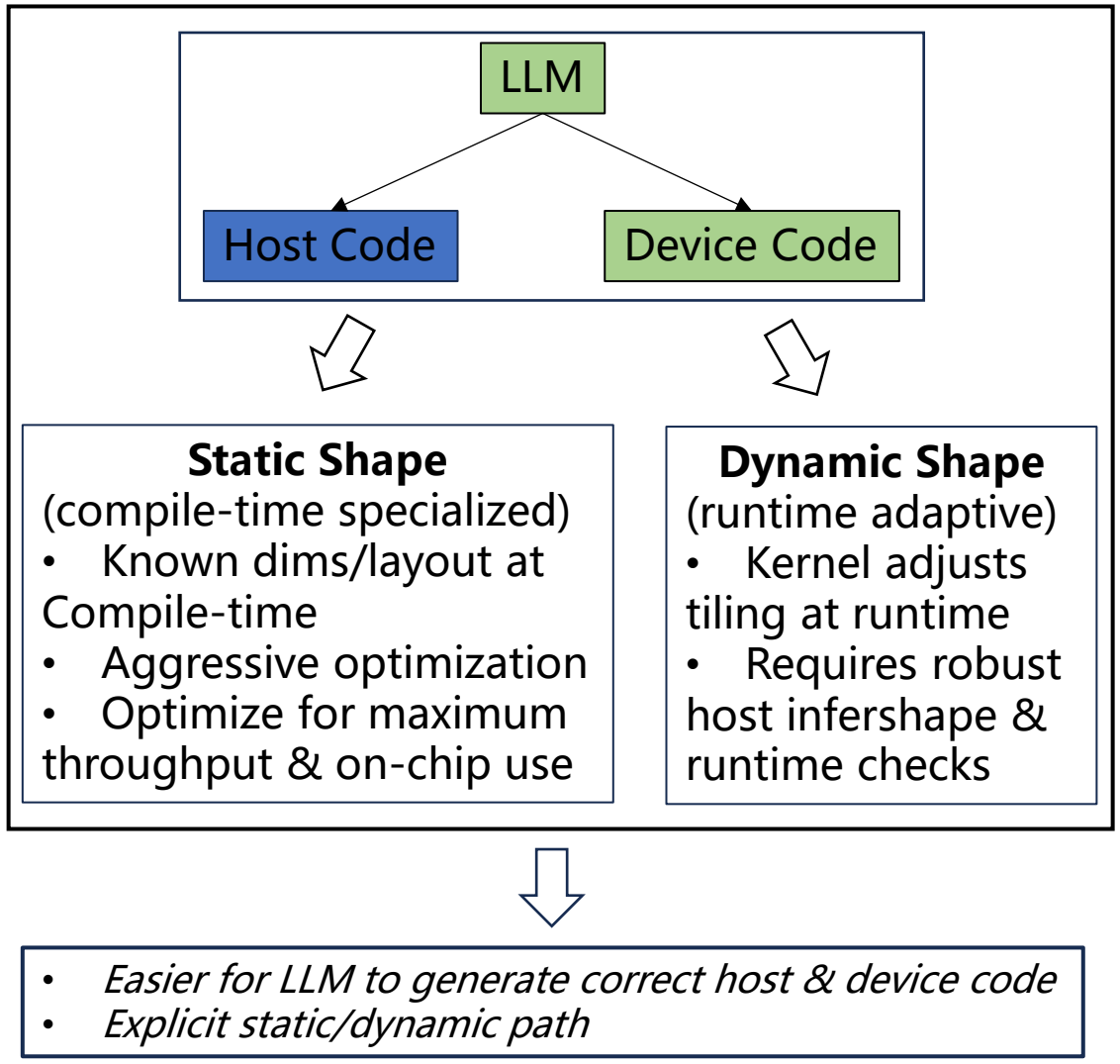
LLM
Host Code
Device Code
Static Shape
(compile-time specialized)
• Known dims/layout at Compile-time
• Aggressive optimization
• Optimize for maximum throughput & on-chip use
Dynamic Shape
(runtime adaptive)
• Kernel adjusts tiling at runtime
• Requires robust host infershape & runtime checks
• Easier for LLM to generate correct host & device code
• Explicit static/dynamic path